Collision energy revisited

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use energy as well as momentum to classify a collision.

SUCCESS CRITERIA - I can...

1. Calculate total momentum before and after.
2. Calculate total kinetic energy independently.
3. Account for reduced kinetic energy without losing total energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “elastic” and “inelastic” to explain your model or result. Sentence starters are optional.

elastic: a collision that conserves total kinetic energy

inelastic: a collision that transforms some kinetic energy to other forms

kinetic energy: energy associated with motion

01 Warm-up: bring back an earlier idea

With right positive, a spring has k=150 N/m and displacement x=-0.05 m.
Find signed spring force and direction.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p_{total,\,i} = p_{total,\,f}$; $K = \frac{1}{2}mv^2$.
Negligible external impulse conserves momentum. Only elastic collisions also conserve total kinetic energy.

Worked reasoning: Two 1 kg carts at +2 and 0 m/s stick. Momentum predicts final speed 1 m/s. Kinetic energy decreases from 2 to 1 J; momentum can be conserved while kinetic energy decreases.

Your turn: Draw separate momentum arrows and kinetic-energy bars before and after.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial | **Transfers** | **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

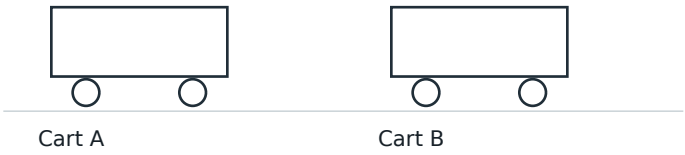
Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Equal-mass carts approach at equal opposite speeds and stick. What happens to total momentum and kinetic energy?

Sentence starter: I notice … . This supports … because … .

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

04 Apply the physics

A 2 kg cart at 3 m/s sticks to a 1 kg cart at rest.
Find v f, K_{before}, K_{after} and energy transformed.
Use the energy model to show the quantities you compare; label units and any unknowns.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

05 Explain your reasoning

Why can a collision be inelastic even when the objects bounce apart?

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A 4 kg cart starts at rest. A 26 N pull and 8 N resistance act over 4 m.
Find final speed.
Show the energy changes and transfers; label units.

Word help: final = at the end of the interval; subscript f

Sentence starter: I found … [unit]. This value means … in this situation.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 1 kg cart at 4 m/s hits an identical stationary cart. Afterward velocities are 1 and 3 m/s.
Check momentum and energy; classify.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 2 kg cart at 4 m/s sticks to a 1 kg cart at -2 m/s.
Find final velocity and fraction of initial kinetic energy lost.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3